

Session 4 · Deck 4.B3

Discretizing an AR(1): Where Π Comes From

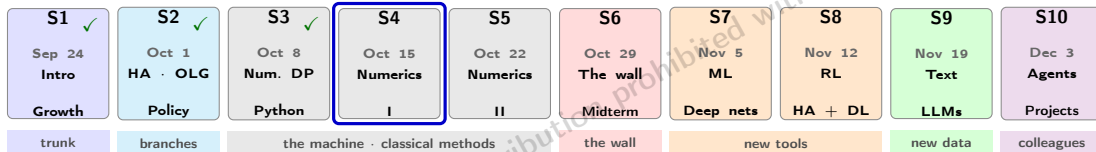
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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

you are here: 4.B3 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 4.B1 approximation and interpolation · 4.B2 quadrature · 4.B3 discretizing an AR(1) · 4.B4 fast and accurate · 4.B5 the worked example, end to end.

A Chain That Reports a Unit Root

Take the standard quarterly productivity process,

$$\ln z' = \rho \ln z + \sigma \varepsilon', \quad \rho = 0.99, \sigma = 0.007,$$

and build a five-state chain the usual way. Then ask the chain what persistence it actually has.

$\rho = 0.99, n = 5$	implied ρ	sd error
Tauchen	1.0000	36%
Rouwenhorst	0.990000	$6 \times 10^{-13}\%$

The Tauchen chain is a **unit root**, and its unconditional standard deviation is a third too small. Every business-cycle moment computed from it is wrong.

Session 3 owed you this

3.A2 said the two standard constructions “disagree badly when $\rho > 0.95$, which is exactly the range macro uses”. This is the measurement.

And $\rho = 0.99$ is not an exotic calibration. It is the **standard quarterly** one — the same ρ used in this session's RBC model and in Homework 1.

What We Are Actually Asking For

A continuous AR(1) has a continuum of states. The Bellman equation needs a finite sum, so we replace it by a chain: a grid $\{z_1, \dots, z_n\}$ and a matrix Π with

$$\Pi_{jm} = \Pr(z' = z_m \mid z = z_j).$$

This is *not* the quadrature problem of 4.B2. There the distribution was fixed; here it **depends on today's state**, and the object we need is a whole conditional law for each j . **What a good chain should match:** the persistence ρ , the unconditional variance $\sigma^2/(1 - \rho^2)$, and ideally the shape of the conditional distribution.

Setup adapted from QM Lec. 4, "AR(1) Discretization: Why?" and "AR(1) Process: Properties".

Why ρ near one is hard

The unconditional standard deviation is

$$\sigma_z = \frac{\sigma}{\sqrt{1 - \rho^2}},$$

which **explodes** as $\rho \rightarrow 1$: at $\rho = 0.99$ it is seven times σ . A fixed number of states has to cover a much wider range, with each step correspondingly coarser.

Tauchen (1986): Put a Grid Down and Integrate the Normal

1. Set $\sigma_z = \sigma / \sqrt{1 - \rho^2}$ and cover $[-m\sigma_z, m\sigma_z]$ with n **equally spaced** points, $m = 3$ by convention.
2. Conditional on z_j , next period is $N(\rho z_j, \sigma^2)$.
3. Assign to each interior z_m the probability that the normal lands in its cell:

$$\Pi_{jm} = \Phi\left(\frac{z_m + \frac{\Delta}{2} - \rho z_j}{\sigma}\right) - \Phi\left(\frac{z_m - \frac{\Delta}{2} - \rho z_j}{\sigma}\right),$$

with the two end states absorbing the remaining tails.

Two things to fix on sight

The formula is often printed twice in the same set of notes, once with a drift term $\mu(1 - \rho)$ and once without, with the state renamed halfway through. **Set $\mu = 0$ and keep one symbol.**

And the coverage multiple is usually called λ — which this course reserves for the invariant distribution. We write m .

The method is intuitive and it is what almost everyone writes first. Its weakness is structural: the grid is chosen *before* the persistence is taken into account.

Tauchen (1986). Algorithm from QM Lec. 4, "Tauchen Method" frames, with the drift inconsistency resolved.

Rouwenhorst (1995): Build the Chain, Not the Integral

A completely different idea: never integrate anything.

Start from the two-state chain

$$\Pi_2 = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}, \quad p = \frac{1+\rho}{2},$$

and grow it recursively. Given Π_{n-1} ,

$$\Pi_n = p \begin{pmatrix} \Pi_{n-1} & 0 \\ 0 & 0 \end{pmatrix} + (1-p) \begin{pmatrix} 0 & \Pi_{n-1} \\ 0 & 0 \end{pmatrix}$$

$$+ (1-p) \begin{pmatrix} 0 & 0 \\ \Pi_{n-1} & 0 \end{pmatrix} + p \begin{pmatrix} 0 & 0 \\ 0 & \Pi_{n-1} \end{pmatrix},$$

then halve the interior rows so they sum to one. The grid is equally spaced on $[-\psi, \psi]$ with $\psi = \sigma_z \sqrt{n-1}$.

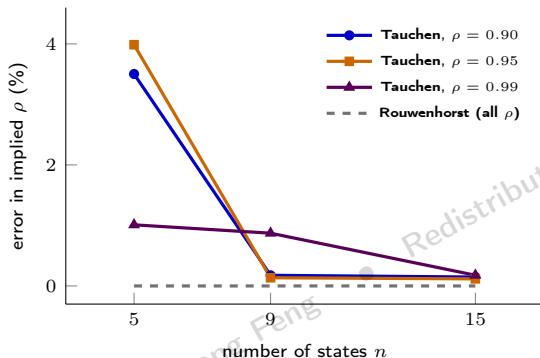
Why it is exact

The construction is a **binomial** one: the chain is a sum of $n-1$ independent two-state processes. Its persistence is $2p-1 = \rho$ and its variance is $\psi^2/(n-1) = \sigma_z^2$ — *by construction*, for every n and every ρ .

The price is that the conditional distribution is binomial, not normal. For a linear-quadratic world that is irrelevant; for a strongly non-linear one it can matter.

Rouwenhorst (1995). Written for this course — the method is named twice in the source decks and never constructed.

The Comparison Session 3 Promised



Error in the chain's implied persistence, as a percentage of the target ρ .

ρ	n	Tauchen	Rouwen.
0.90	5	3.50%	2×10^{-14}
0.95	5	3.99%	0
0.99	5	1.01%	0
0.99	15	0.18%	7×10^{-14}

Rouwenhorst is exact to machine precision at **every** n and **every** ρ . The two lines are not close; one of them is zero.

And the Variance Is Worse Than the Persistence

Persistence is the forgiving statistic. The unconditional standard deviation — which is what sets the *size* of every business-cycle moment — is where Tauchen really suffers.

sd error		Tauchen	Rouwenhorst
$\rho = 0.90$	$n = 5$	26.9%	≈ 0
$\rho = 0.95$	$n = 5$	31.4%	≈ 0
$\rho = 0.99$	$n = 5$	36.2%	≈ 0
$\rho = 0.99$	$n = 9$	28.5%	≈ 0
$\rho = 0.99$	$n = 15$	20.1%	≈ 0

Read the last row

Even with **fifteen** states, at the standard quarterly persistence, the Tauchen chain's unconditional standard deviation is **a fifth too small**.

A model calibrated with that chain will understate the volatility of output — and the natural response, raising σ until the moments match, hides a discretization error inside a structural parameter.

Target $\sigma_z = \sigma / \sqrt{1 - \rho^2}$; chain moments computed from the invariant distribution of each Π .

What To Actually Do

Method	Use it when
Rouwenhorst	the default , and essential for $\rho > 0.9$
Tauchen	low persistence, or when the conditional distribution must look normal
Tauchen–Hussey	quadrature-based; accurate for moderate ρ , degrades badly near one

Kopecky & Suen (2010) make the case formally:

Rouwenhorst matches the conditional and unconditional first and second moments and the persistence *exactly*, for any n .

Kopecky & Suen, "Finite State Markov-Chain Approximations to Highly Persistent Processes", *RED* (2010).

The habit to build

Whatever you use, **report what the chain actually delivers**: its implied ρ and σ_z next to the targets. It is three lines of code and it is the only way anyone can tell whether your moments came from the economics or the grid.

Homework 1 Part B asks for exactly that table at $n = 5, 9, 15$ — and then asks you to solve the model with both and see whether it moved the answer.

Takeaways

- Π is not data and it is not a modelling choice you can leave implicit. It is an **approximation with an error**, and the error is measurable.
- **Tauchen** puts down an equally spaced grid and integrates the conditional normal over each cell. Intuitive, and the grid is chosen before persistence is taken into account.
- **Rouwenhorst** builds the chain recursively from a two-state block and matches ρ and σ_z **exactly, by construction**, for every n .
- Measured at the standard quarterly calibration $\rho = 0.99$: the five-state Tauchen chain implies $\rho = 1.0000$ — **a unit root** — and a standard deviation 36% too small. At $n = 15$ it is still 20% too small.
- **Use Rouwenhorst**, and report the chain's implied ρ and σ_z beside the targets every time.

Bridge: Everything Is Now in Place

Session 3 ended with six crude choices and a list of what would repair each. We have now built all of them: a root-finder, a maximizer on an interval, derivatives, an interpolation scheme that keeps its shape, a quadrature rule, and a transition matrix that is not invented.

What we have not done is put them back together. The crude solver of 3.A4 took 1,352 sweeps and 2.4 billion comparisons at this session's calibration, and its worst Euler error was $10^{-1.5}$. The question for the last deck is simple: **how much of that was avoidable, and which repair bought the most?**

Next (4.B4)

Howard's policy improvement, monotonicity, concavity — and the endogenous grid method, done the way Carroll actually formulated it, which removes the root-finding entirely.

Then one table: every technique, its speed-up and its accuracy, measured on the same model.